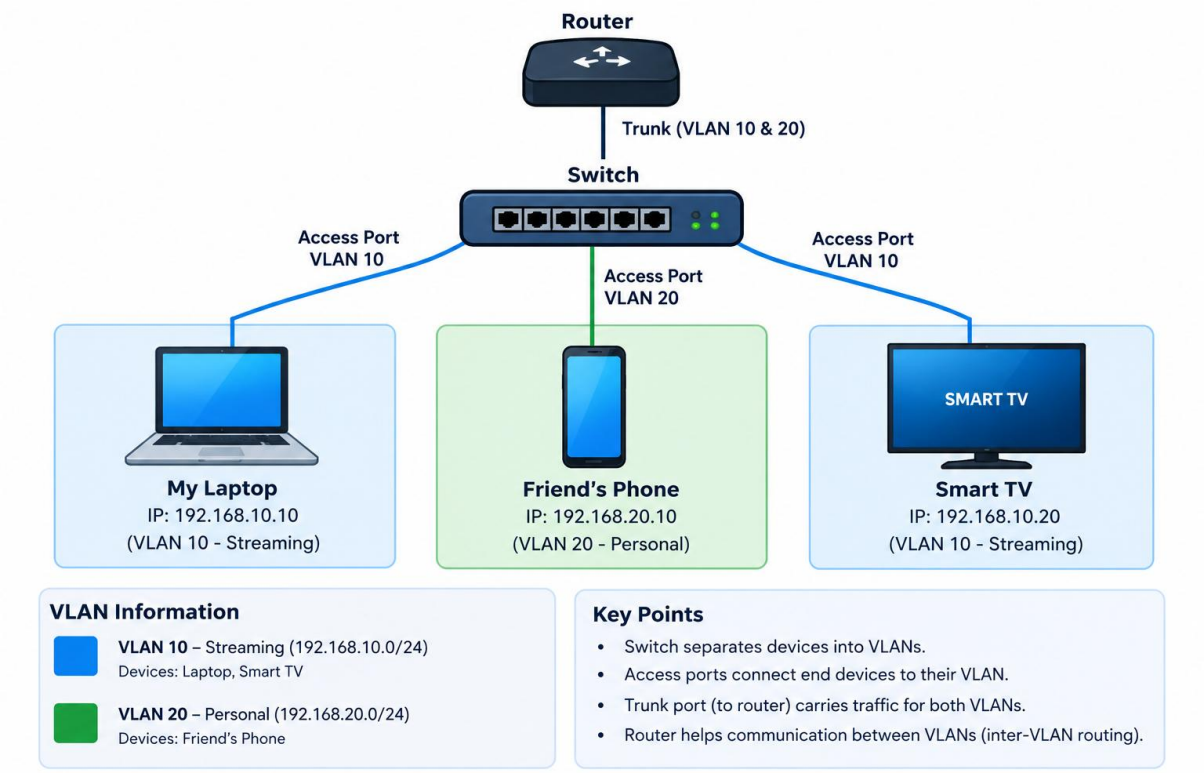
SWITCHING AND VLANS

1. Draw a simple network diagram showing three devices (like your laptop, a friend's phone, and a smart TV) connected to a switch, and label which device is in which VLAN (e.g., VLAN 10 for streaming, VLAN 20 for personal devices).



2. Configure a simulated switch using Cisco Packet Tracer or NetSim: create two VLANs (VLAN 10 and VLAN 20), assign three PCs to VLAN 10 and two PCs to VLAN 20, and verify that PCs in different VLANs cannot ping each other.

Hint: Use the 'vlan' and 'switchport access vlan' commands in the simulator CLI.

VLAN Configuration in Cisco Packet Tracer

1. Create VLANs and assign ports:

Switch> enable

Switch# configure terminal

Switch(config)# vlan 10

Switch(config-vlan)# name Streaming

Switch(config)# vlan 20

Switch(config-vlan)# name Personal

Switch(config)# interface range fa0/1-3

Switch(config-if-range)# switchport mode access

Switch(config-if-range)# switchport access vlan 10

Switch(config)# interface range fa0/4-5

Switch(config-if-range)# switchport mode access

Switch(config-if-range)# switchport access vlan 20

Switch(config)# end

2. PC Assignment:

- VLAN 10: PC1, PC2, PC3
- VLAN 20: PC4, PC5

3. Verify VLANs:

Switch# show vlan brief

4. Test connectivity:

- Ping between PCs in VLAN 10: Successful (if IPs are configured correctly).
- Ping between VLAN 10 and VLAN 20: Failed, because inter-VLAN routing is not configured.

3. In any network simulator, set up a scenario where you move a PC from VLAN 10 to VLAN 20. Show the before and after configuration commands and test connectivity to confirm the change.

Move PC from VLAN 10 to VLAN 20

1. Before Configuration (VLAN 10)

Switch> enable

Switch# configure terminal

Switch(config)# interface fa0/1

Switch(config-if)# switchport mode access

Switch(config-if)# switchport access vlan 10

2. After Configuration (VLAN 20)

Switch(config)# interface fa0/1

Switch(config-if)# switchport access vlan 20

Switch(config-if)# end

Switch# show vlan brief

3. Test Connectivity

- Ping a PC in VLAN 10: Fails (assuming no inter-VLAN routing).
- Ping a PC in VLAN 20: Succeeds if IP addresses and subnet settings are correct.

4. Write a short script or command list that would automate assigning 10 ports on a switch to VLAN 30, using the syntax of your chosen simulator (Packet Tracer, NetSim, or GNS3).

Constraint: Use a loop or range command if supported by the tool to avoid repeating the same command 10 times.

Automate VLAN 30 Assignment (Cisco Packet Tracer)

Switch> enable

Switch# configure terminal

Switch(config)# vlan 30

Switch(config-vlan)# exit

Switch(config)# interface range fa0/1-10

Switch(config-if-range)# switchport mode access

Switch(config-if-range)# switchport access vlan 30

Switch(config-if-range)# end

Switch# show vlan brief